**Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.**

Functions of the Components

- The CPU (The Processing Core): it is the brain of the computer, It performs all critical mathematical calculations, logical operations, and data processing. The CPU interacts directly only with high-speed system memory (RAM) or Graphics cards via dedicated lanes. It relies on the chipset to communicate with the rest of the computer.
- The Chipset (Platform controller Hub): Built into the motherboard, the chipset acts as the central traffic controller and translator for the system. It manages data flow between the CPU and all peripheral devices—such as storage drives (SSDs/HDDs), USB ports, network adapters (Wi-Fi/Ethernet), and audio hardware.

Workflow:
When a user executes a command, such as opening an application like Spotify,
Step 1: When we double-click the spotify application icon, the input signal is sent from the mouse via USB or Bluetooth to the chipset. This hardware signal is converted into system data, and sends a request to the CPU.
Step 2: The CPU then instructs the chipset to retrieve the stored files of the application. The chipset accesses the hard drive (HDD/SSD), extracts the necessary application data, and loads it into the system memory (RAM) where the CPU can rapidly access it.
Step 3: Once the data is in the RAM, the CPU processes the software instructions from the RAM
Step 4: When the user selects a song, the CPU sends a signal to the chipset to use the network card (Wi-Fi) to stream data from the external servers. The chipset fetches the incoming audio packets and delivers them to the CPU. The CPU decodes the compressed digital audio file and sends the raw audio stream back to the chipset, which routes it directly to the motherboard's onboard audio codec to be converted into sound waves through the speakers or headphones.

**Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.**